



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	INDUSTRIAL CATALYSIS				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic understanding of surface chemistry 2. A sound knowledge in reaction kinetics and thermodynamics				
Course Summary	The catalysis course covers catalyst preparation, deactivation, and applications across various fields. It includes fundamental principles, preparation methods, deactivation mechanisms, and regeneration techniques. Additionally, it explores phase transfer catalysis, biocatalysis, and industrial catalysis, focusing on their principles, mechanisms, and applications such as oil-based chemistry, hydrocarbon synthesis, and environmental protection. Ultimately, students gain a comprehensive understanding of catalytic processes and their applications in chemistry, environmental science, and materials science.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental concepts in catalysis, including catalyst preparation, mechanisms of catalytic reactions, and catalyst deactivation processes.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Apply various preparative methods, such as phase transfer catalysis, biocatalysis, and industrial catalysis, in the synthesis and transformation of chemical compounds.	Ap	C	Class test /Assignment /Quiz

CO3	Analyze and classify catalyst deactivation processes, including poisoning, coke formation, and sintering, and explore methods for catalyst regeneration.	An	C	Assignment/ Class test
CO4	Evaluate the principles and applications of industrial catalytic processes in oil-based chemistry, hydrocarbon synthesis, environmental protection, and polymerization reactions.	E	P	Assignments /Seminar presentation
CO5	Explore emerging catalytic technologies such as biodiesel production, photocatalysis, and electrocatalysis, and understand their potential applications and challenges in contemporary industries and research fields.	E	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Catalyst Preparative Methods and Deactivation		12	20
	1	Support materials -Preparation and structure of supports, Surface properties.	2	
	2	Preparation of catalysts- Introduction of precursor compound, Pre-activation treatment, Activation process.	2	
	3	General methods of synthesis of zeolites. Mechanism of nuclear formation and crystal growth. Structures of some selected zeolites. Zeolites A, X and Y. Shape selective catalysis.	3	
	4	Deactivation of catalysts- Classification of catalyst deactivation processes, Poisoning of catalysts, Coke formation on catalysts, Sintering of catalysts.	3	

	5	Regeneration of deactivated catalysts. Feasibility of regeneration. Description of coke deposit and kinetics of regeneration	2	
II	Phase Transfer Catalysis		12	20
	6	Basic concepts in phase transfer catalysis. Phase transfer catalyzed reactions and their basic steps.	2	
	7	Effect of reaction variables on transfer and intrinsic rates. Outline of compounds used as phase transfer catalysts. Use of quaternary salts.	3	
	8	Macrocyclic and macrobicyclic ligands. PEG's and related compounds.	3	
	9	Use of dual phase transfer catalyst or co-catalyst in phase transfer systems.	2	
	10	Separation and recovery of phase transfer catalysts. Insoluble phase transfer catalysts.	2	
III	Biocatalysis		12	20
	11	Enzymes. An introduction to enzymes. Enzymes as proteins.	2	
	12	Classification and nomenclature of enzymes. Structure of enzymes. Working of enzymes. Effect on reaction rate. Thermodynamic definitions.	2	
	13	Catalytic power and specificity of enzymes. Optimization of weak interactions between enzyme and substrate in the transition state.	3	
	14	Binding energy, reaction specificity and catalysis. Specific catalytic groups contributing to catalysis. Immobilized biocatalysts.	3	
	15	Definition and classification of immobilized biocatalysts. Immobilization of coenzymes.	2	
IV	Industrial Catalysis		12	20
	16	Oil based chemistry- Catalytic reforming, Catalytic cracking, Paraffin cracking, Steam cracking.	1	
	17	Hydrocarbons from synthesis gas. Fisher-Tropsch process. Mobil process for conversion of methanol to gasoline hydrocarbons.	2	
	18	Catalysis for environmental protection, removal of pollutants from exhausts, mobile and static sources.	2	

	19	Hydroformylation of olefins. Carbonylation of organic substrates.	2	
	20	Conversion of methanol to acetic acid. Synthesis of vinyl acetate and acetic anhydride. Palladium catalyzed oxidation of ethylene.	3	
	21	Acrylonitrile synthesis. Zeigler-Natta catalysts for olefin polymerization.	1	
	22	Propene polymerization with silica supported metallocene/MAO catalysts	1	
V	Open Ended Module		12	20
	Open Ended	Biodiesel via catalytic process, Photocatalysis, Electrocatalysis, A survey of important Indian catalytic industries and their products. Experiments involving preparation of catalysts and catalytic reactions.	12	

References:

1. J.R. Anderson and M. Boudart (Eds), "Catalysis, Science and Technology", Vol 6, Springer- Verlag, Berlin Heildberg, 1984.
2. R.B. Anderson, "Experimental methods in catalysis research", Vol I, II, Academic press, NY, 1981.
3. R. Szostak, "Molecular sieves: principles of synthesis and identification", Van Nostrand, NY, 1989.
4. R. Hughes, "Deactivation of catalysts", Academic press, London, 1984.
5. C.M. Starks, C.L. Liotta and M. Halpern, "Phase Transfer Catalysis – Fundamentals, Applications and Industrial Perspectives", Chapman & Hall, New York, 1994.
6. A.L. Lehninger, "Principles of Biochemistry", Worth Publishers, USA, 1987.
7. G. Ertl, H. Knozinger and J. Weitkamp, "Handbook of Heterogeneous Catalysis" Vol 1- 5, Wiley-VCH, Weinheim, 1997.
8. R.J. Farrauto and C.H. Bartholomew, "Fundamentals of Industrial Catalytic Processes", Blackie Academic and Professional – Chapman and Hall, 1997.
9. R. Pearce and W.R. Patterson, "Catalysis and chemical processes", Academic press, Leonard Hill, London, 1981

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3		1		1		
CO 2	2				1	2	3		1		2		1
CO 3						1	3		2		2		1
CO 4				2		1	3		2		2		2
CO 5				2	1	3	3		2		2		2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓